Special in Repo, Thin on the Ask

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Abstract

I document how the two sides of the interdealer cash order book of Italian government bonds vary with repo specialness, using the best quotes, the full order book rebuilt from quote records, and executed orders. With bond and day fixed effects, a bond with 2 bp of specialness has 7.99 log points less depth at the best ask than a bond at the general collateral rate, while its quoted spread does not differ. This difference holds within bond-months and with controls for the bond's life cycle. Over longer horizons, special bonds also have more depth at the best bid and fewer executed sells relative to buys. Special bonds have less best-ask depth whether their repo borrowing is above or below its recent average. In my model of dealer quotes, a cost of sourcing a bond that is hard to borrow raises dealers' reservation prices, and with discrete prices dealers then withdraw from the best ask and join the best bid.

Keywords: Repo specialness, Limit order book, Market depth, Market making, Short selling, Sovereign bonds

JEL classification: E43, G12, G14, G24

1 Introduction

The repo rate on a specific government bond is the price of borrowing that bond, and the amount by which it falls below the general collateral (GC) rate when the bond is scarce is called specialness (Duffie, 1996; Jordan and Jordan, 1997). A bond becomes special when the demand to borrow it, for example from dealers who have sold it short, exceeds the supply that its holders are willing to lend. Its owner can then borrow cash against it at a lower rate, so a bond that is expected to be special is worth more. Specialness rises around auctions and

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for recently issued bonds (Krishnamurthy, 2002; Graveline and McBrady, 2011), and Corradin and Maddaloni (2020) and Arrata et al. (2020) find that in the euro area it rose when the Eurosystem's asset purchases reduced the supply of bonds available for lending. Scarcity in the repo market matters for the cash market because dealers supply the quotes in the cash market, and a dealer who sells a bond that it does not hold must source the bond for delivery. This paper documents how the two sides of the interdealer cash order book vary with specialness. To interpret these patterns, I develop a model of dealer quotes in which sourcing a scarce bond is costly.

I start from the standard inventory model of a dealer, in which the dealer's reservation price, the price at which it is indifferent to trading a marginal unit, falls with the inventory it holds (Stoll, 1978; Ho and Stoll, 1981; Avellaneda and Stoikov, 2008). I add one friction: once a bond is hard to borrow, a dealer that will have to deliver more than it holds must source the shortfall at a cost. Each unit the dealer holds saves that cost, so the cost raises the dealer's reservation price. As in models of the limit order book with several price levels, prices are discrete, and a quote at the best price executes more often but earns less per unit (Goettler et al., 2005; Hollifield et al., 2004). A dealer whose reservation price rises past a cutoff moves its ask behind the best price or its bid to the best price, while the ask quotes that stay at their price shrink and the bid quotes that stay grow. The best ask and bid prices themselves do not change as long as at least one dealer keeps quoting at each of them. The model therefore predicts that specialness reduces depth at the best ask and increases depth at the best bid without changing the quoted spread. It also predicts that aggressive sell orders become less frequent relative to aggressive buy orders, and that these effects level off once specialness is high enough for every dealer to bear the cost.

I test these predictions on Italian government bonds on MTS from September 2019 to February 2021, using the best quotes averaged over 15-minute intervals, the full order book rebuilt from the exchange's quote records, and executed orders with their aggressor side. With bond and day fixed effects, a bond with 2 basis points (bp) of specialness has 7.99 log points less depth at the best ask than a bond at the GC rate and 8.41 log points more depth at the best bid. Its quoted spread does not differ. In the full order book, the difference on the ask side is concentrated at the best price, and depth behind the best ask does not differ significantly. The probability of an executed sell is 8.73 percentage points lower and that of an executed buy 3.64 percentage points higher. The ratio of best-ask to best-bid depth and executed selling change

with specialness up to about 2 bp and much more slowly beyond.

The main finding is thus a difference in where liquidity is quoted: special bonds have less depth at the best ask and more at the best bid, while their quoted spread and total depth barely differ. Liquidity measures based on the quoted spread or on total depth can therefore miss this difference. A buy of €25 million is 2.65 percentage points less likely to be filled in full at the best ask of a special bond, and a sell of the same size is 3.33 percentage points more likely to be filled in full at its best bid.

The lower best-ask depth of special bonds persists with bond-by-month fixed effects, with controls for the bond's life cycle and quoting obligations, and with specialness cleaned of the repo bid-ask bounce. It also persists without auction windows or futures delivery windows and on bond-days without net buying on MTS over the previous five days. Following individual quotes within the day, fewer quotes move to the best ask of a special bond, while more quotes move to its best bid and fewer leave the book from it. The ask/bid depth ratio also falls within bond-weeks and on the day specialness changes, whereas the bid and selling results hold only over longer horizons. Finally, repo borrowing above its recent average does not reduce the best-ask depth of a bond that trades at the GC rate, whereas special bonds have less best-ask depth whatever their repo borrowing.

The paper contributes to three literatures. First, the literature on repo specialness, from Duffie (1996) to Arrata et al. (2020), explains why bonds become special and how specialness enters bond prices. I show how each side of the cash order book varies with specialness. Second, work on short-selling frictions finds that costly or banned short selling reduces liquidity, mostly in equities and around short-selling bans.[1] I study instead a continuous friction in an interdealer bond market, on each side of the book and at each price level. Third, models of market making explain quotes through dealer inventory and order placement (Ho and Stoll, 1981; Avellaneda and Stoikov, 2008; Roşu, 2009), and with discrete prices liquidity suppliers adjust the depth they display as well as their prices (Harris, 1994; Kavajecz, 1999; Goldstein and Kavajecz, 2000). In my model, a sourcing cost that is small relative to the tick changes depth at the best prices and leaves the prices themselves unchanged, because prices are discrete.

Section 2 reviews the literature, Section 3 presents the model, and Section 4 describes the data and the design. Section 5 reports the results, and Section 6 concludes.

[1] These include the ban of the US Securities and Exchange Commission on short sales of nearly 1,000 financial stocks from 19 September to 8 October 2008, which Boehmer et al. (2013) study, and the bans imposed around the world during the crisis of 2007 to 2009, which Beber and Pagano (2013) study in a sample of 30 countries.

2 Related literature

Repo specialness. Duffie (1996) shows that specialness arises when the demand to borrow a particular bond exceeds the supply of it for lending. Demand to short a bond rises around auctions and for recently issued bonds (Jordan and Jordan, 1997; Krishnamurthy, 2002; Graveline and McBrady, 2011), and Vayanos and Weill (2008) show how search frictions concentrate it in a few bonds. Supply falls when central banks buy bonds (D'Amico et al., 2018; Corradin and Maddaloni, 2020; Arrata et al., 2020). Specialness is also priced, because the owner of a bond that is expected to be special can lend it out (Duffie, 1996; Duffie et al., 2002; D'Amico and Pancost, 2022), and Buraschi and Menini (2002) find that term repo rates on special bonds embed a liquidity risk premium. In the euro area, Jank et al. (2021) link specialness to the scarcity of safe assets and to collateral reuse, Schaffner et al. (2019) document how repo rates differ across collateral, and Greppmair and Jank (2026) show that securities lending by the Eurosystem makes scarce bonds less special and their cash market more liquid. For Italian government bonds on MTS, Dufour et al. (2020) explain specialness with the quantity offered at the best ask prices, with order flow and with the auction cycle, and MEF, MTS and CRIEP (2024) report that bid-ask spreads and specialness predict each other. These studies use the book to explain specialness or link specialness to the bid-ask spread. I instead examine how each side of the book, depth at and behind the best price, and executed orders vary with specialness, and I interpret these differences with a model of dealer quotes.

Short-selling frictions and liquidity. Borrowing a security costs more than the lending fee. D'Avolio (2002) shows that lenders can recall a loan. Borrowers must also search for lenders (Duffie et al., 2002; Kolasinski et al., 2013), and Engelberg et al. (2018) find that the risk that fees rise or loans are recalled deters short selling. An options market maker that finds a stock hard to borrow often chooses to fail to deliver it instead of borrowing it (Evans et al., 2009). In Italian government bonds, failed deliveries became more likely as bonds became more special (Corradin and Maddaloni, 2020). Diamond and Verrecchia (1987) show that short-sale constraints slow the adjustment of prices to bad news. The short-selling bans of 2008 reduced liquidity and widened bid-ask spreads (Beber and Pagano, 2013; Boehmer et al., 2013), and Cenesizoglu and Grass (2018) find that in the NYSE limit order book the ask side lost liquidity during the 2008 ban while the bid side did not. When market makers have market power, short-sale constraints lower bid prices and raise ask prices (Liu and Wang, 2019), and Comerton-Forde et al. (2016)

show that short sales also come from liquidity providers. Most of this work concerns equities and discrete events such as bans.

Market making and the limit order book. The model builds on two strands of market microstructure. In inventory models, a dealer quotes around a reservation price that falls with the inventory it holds (Stoll, 1978; Amihud and Mendelson, 1980; Ho and Stoll, 1981). In continuous time, a dealer who is short raises both quotes (Avellaneda and Stoikov, 2008), and with an inventory limit it stops selling once it reaches the limit (Guéant et al., 2013). Comerton-Forde et al. (2010) and Hendershott and Menkveld (2014) find that market-maker inventories move liquidity and prices. In models of the limit order book, traders weigh a better price against a lower or later chance of execution (Glosten, 1994; Parlour, 1998; Foucault, 1999; Foucault et al., 2005; Goettler et al., 2005; Roşu, 2009). In the model of Hollifield et al. (2004), where a trader places an order depends on how much it values the asset. Their tests on Stockholm order data support this for buy orders and for sell orders taken separately. Sandås (2001) tests such models against depth at several levels of the book. Because prices are discrete, liquidity suppliers also choose how much depth to display at a price. A larger tick raises quoted depth (Harris, 1994; Goldstein and Kavajecz, 2000), and Kavajecz (1999) finds that specialists reduce their quoted depth when the risk of trading with informed traders rises. I combine the two strands: the sourcing cost enters the dealer's valuation like a short position, and price discreteness turns the change in valuation into a change in depth at the best prices.

Repo and cash markets through dealers. Closest to my mechanism, Huh and Infante (2021) model dealers who source bonds in the repo market to serve their clients, so that specialness raises the bid-ask spread that dealers charge. Valseth (2025) finds that Norwegian primary dealers borrow more bonds in repo when their net passive sales in the limit order book rise, and that bond market liquidity worsens when repo specialness is high. My model differs in where the cost falls: because it falls on the dealer's own sales, it also raises the value of holding the bond, so depth moves from the ask to the bid while the best prices need not change. On MTS, Pelizzon et al. (2016) study how liquidity varies with credit risk and central bank intervention, O'Sullivan and Papavassiliou (2020) how it varies with maturity, and Dunne et al. (2015) study the link between the interdealer and customer markets. I add depth by side and by level of the book.

3 Model

The model starts from the standard inventory model of a dealer and adds one friction. A dealer that sells a bond must deliver it, and sourcing a bond that has become hard to borrow is costly. I then let the dealers quote at discrete prices, as in models of the limit order book. The model is static and partial equilibrium: it takes the quoted prices and the execution probabilities as given and does not solve for equilibrium prices. Appendix A1 of the online appendix gives the proofs.

3.1 Dealers' valuation

Dealer i values an inventory of y units of the bond at

$$W_i(y) = v\,y - \frac{\gamma}{2}\,y^2 - k_i\,\mathrm{E}[(D_i - y)^+]. \tag{1}$$

The first two terms are the standard inventory model, in which the bond is worth v and $\gamma > 0$ is the cost of inventory risk. In the third term, which is new, the random quantity $D_i \geq 0$ is what the dealer will have to deliver later, for example to clients or against its own later sales. A dealer that holds less than D_i must source the shortfall at k_i per unit.

The direct cost of borrowing a special bond is small: at 2 bp of specialness, borrowing €10 million for one night costs about €5.56. Sourcing a bond that has become hard to borrow costs more, because the cost covers locating the bond and the risks of a recall and of a failed delivery (D'Avolio, 2002; Kolasinski et al., 2013; Corradin and Maddaloni, 2020). I treat this cost as a reduced-form friction that applies once specialness S exceeds a threshold $s_i^* \geq 0$, which differs across dealers:

$$k_i = k\,\mathbf{1}[S > s_i^*]. \tag{2}$$

Below zero, the bond's special collateral (SC) repo rate lies above the GC rate, so the bond is not hard to borrow and $k_i = 0$.

The dealer's reservation price, the price at which it is indifferent to trading a marginal unit, is the value of one more unit of inventory,

$$W_i'(y) = v - \gamma\,y + k_i\,\Pr(D_i > y). \tag{3}$$

The first two terms are the reservation price of the standard model, which falls with the inventory

the dealer holds (Ho and Stoll, 1981; Avellaneda and Stoikov, 2008). The sourcing cost adds to it, because each unit the dealer holds saves sourcing on a later delivery. Market makers hold little of a bond relative to what they deliver, so I assume that a dealer's later deliveries exceed any inventory its quotes can lead to.

Assumption 1. With probability one, $D_i > \max\{y_i,\, (v + k - b_2)/\gamma\}$, where y_i is the dealer's inventory and b_2 is the lowest bid price in Section 3.2.

Lemma 1 (Reservation price)**.** *Under Assumption 1,* $W_i(y) = (v + k_i)\, y - \gamma y^2/2 - k_i\, \mathrm{E}[D_i]$ *over the range of the dealer's quotes. The sourcing cost raises the dealer's reservation price from* $r_i = v - \gamma y_i$ *to* $r_i + k_i$*. Its effect equals that of a fall of* k_i/γ *in the dealer's inventory.*

Lemma 1 says that the cost works like a short position. Without Assumption 1, the cost raises the value of a marginal unit by $k_i \Pr(D_i > x)$, which depends on the inventory x after a trade, so the responses below would depend on the distribution of D_i. Assumption 1 keeps the rise the same at every inventory the quotes can reach.

3.2 Quotes at discrete prices

On each side of the book a quote stands at the best price or one tick behind it. Ask prices are a_1 and $a_2 = a_1 + \Delta$, and bid prices b_1 and $b_2 = b_1 - \Delta$, where Δ is the tick and level 1 is the best price. A continuum of dealers quotes in the market, and their reservation prices r_i for a bond at the GC rate have density h. The prices are set by the market, so a single dealer takes them as given. Section 4.2 shows that quoting obligations on MTS do not fix where a quote stands relative to the best price. Each dealer posts one two-sided quote and chooses the level and the size on each side, and a quote at level j executes in full with probability π_j on either side, with $\pi_1 > \pi_2 > 0$.

Assumption 2. At most one side of a dealer's quote executes before the dealer revises the quote.

In continuous-time inventory models, orders arrive one at a time and the dealer can revise its quotes after each trade (Ho and Stoll, 1981; Avellaneda and Stoikov, 2008). Assumption 2 lets the dealer choose each side of its quote on its own, and it requires the execution probabilities of the two sides to sum to at most one, which holds if $\pi_1 \leq 1/2$. If both sides can execute, the size responses below keep their signs but are smaller (Appendix A1).

Lemma 2 (Quote size)**.** *If a dealer with reservation price r quotes at level j, it quotes q_j^a on the ask and q_j^b on the bid, with expected gains V_j^a and V_j^b:*

$$q_j^a = \frac{(a_j - r)^+}{\gamma}, \quad V_j^a = \frac{\pi_j\left[(a_j - r)^+\right]^2}{2\gamma}, \qquad q_j^b = \frac{(r - b_j)^+}{\gamma}, \quad V_j^b = \frac{\pi_j\left[(r - b_j)^+\right]^2}{2\gamma}. \tag{4}$$

A quote is larger when its price lies further from the dealer's reservation price and smaller when inventory risk is costlier. The execution probability scales the gain but not the size, because inventory risk arises only if the quote executes.

Lemma 3 (Price level)**.** *A dealer with reservation price r places its ask at the best price if and only if $r < \bar{r}^a$, and its bid at the best price if and only if $r > \bar{r}^b$, where*

$$\bar{r}^a = a_1 - \frac{\sqrt{\pi_2}}{\sqrt{\pi_1} - \sqrt{\pi_2}}\,\Delta, \qquad \bar{r}^b = b_1 + \frac{\sqrt{\pi_2}}{\sqrt{\pi_1} - \sqrt{\pi_2}}\,\Delta. \tag{5}$$

The best price executes more often but earns less per unit. A dealer that values the bond below $\bar{r}^a$ therefore earns enough at the best ask to prefer it, and a dealer that values the bond above $\bar{r}^b$ prefers the best bid. Where a dealer quotes thus depends on its reservation price, as in order-choice models of the limit order book (Hollifield et al., 2004; Goettler et al., 2005).

3.3 Specialness and the book

Compare a bond at the GC rate with a bond whose specialness exceeds the thresholds of some dealers. The thresholds are independent of the dealers' inventories, so a share $G(S)$ of the dealers at every reservation price is affected, where G is the distribution of the thresholds. By Lemma 1, the reservation price of each affected dealer rises by k.

Proposition 1 (The book)**.** *Let specialness rise from zero to S.*

(a) Ask quotes of affected dealers that stay at their price shrink by k/γ. Affected dealers with $r_i < \bar{r}^a \leq r_i + k$ move their ask from the best price to the price behind it. Depth at the best ask falls, because fewer quotes stand there and the quotes that stay are smaller.

(b) Bid quotes of affected dealers that stay at their price grow by k/γ. Affected dealers with $r_i \leq \bar{r}^b < r_i + k$ move their bid from behind the best price to the best price. Depth at the best bid rises, because more quotes stand there and the quotes that stay are larger.

(c) Suppose that h is non-increasing on $[\bar{r}^a - k,\, a_2)$ and non-decreasing on $(b_2 - k,\, \bar{r}^b]$. Then depth behind the best ask does not fall, and depth behind the best bid does not rise. Depth at the best ask therefore falls by more, in proportion, than depth behind it, and depth at the best bid rises by more, in proportion, than depth behind it.

(d) Total depth on each side can rise or fall. The best ask stays at a_1 as long as a dealer that does not move still quotes it, and the best bid stays at b_1 in the same way, so the quoted spread and the mid-quote do not change. If every dealer leaves the best ask, the best ask moves to a_2, the quoted spread widens by one tick and the mid-quote rises by half a tick.

Parts (a) and (b) describe two ways in which the book responds: dealers that keep their quote at the same price change its size, and dealers near the cutoffs move their quote to the other price. The change in size is the same for every dealer that stays, whatever its inventory, because inventory sets the level and the size of the dealer's quote but not the change that the cost causes. The model does not sign the average size of the quotes at the best price, because the quotes that leave the best ask, and those that join the best bid, are the smallest there. Part (c) needs a condition because quotes behind the best ask shrink while the quotes that move there add depth. After the cost, the dealers that quote behind the best ask are those whose reservation price was up to k lower. If the density of reservation prices does not rise towards a_2, there are at least as many of them, and depth behind the best ask does not fall. The bid side is the mirror image. The condition holds when the dealers' reservation prices have a single peak near the mid-quote and the best prices lie several ticks from it. This is the case on MTS, where the median quoted spread is 14.07 bp of price (Table 1) and the tick is 1 bp for most bonds (Section 4.2). Part (d) explains why the quoted spread can stay the same: the best ask moves only if every dealer there leaves it, which in the model requires every dealer to bear the cost.

Because the execution probabilities are fixed, the model leaves out a feedback. More aggressive buying would raise the chance that an ask at the best price executes and draw some quotes back to it, and fewer aggressive sells would lower that chance at the best bid. The predictions hold as long as this feedback is weaker than the direct effect of the cost. A dealer could also bid one tick above the best bid, which would raise the best bid and the mid-quote, but this pays only if the cost is large relative to the tick, and I assume that it is not. With more price levels behind the best price, quotes that move between them grow on the ask and shrink on the bid, which strengthens part (c). The minimum quote size on MTS can remove small quotes behind

the best price, which works against it.

Proposition 2 (Aggressive orders)**.** *A dealer with reservation price r sells aggressively at the best bid if and only if $b_1 - r > c$, and buys aggressively at the best ask if and only if $r - a_1 > c$, where $c > 0$ is the cost per unit of an aggressive order. A rise in reservation prices therefore lowers the share of dealers that sell aggressively and raises the share that buy aggressively.*

Proposition 3 (Shape)**.** *Every outcome of Propositions 1 and 2 depends on specialness only through the set of affected dealers. It therefore stops changing once S exceeds every threshold. While the best prices do not change, the expected change in depth and in the number of aggressive orders equals $G(S)$ times the change when every dealer is affected. It is linear in S up to the largest threshold if and only if G is uniform there. Logs and ratios of these quantities are not proportional to $G(S)$, but they are monotone in S.*

Proposition 3 assumes that k does not rise further with specialness. If it did, for example because failed deliveries become more likely as a bond becomes more special (Corradin and Maddaloni, 2020), the responses would keep growing beyond the thresholds.

3.4 Predictions

Propositions 1 to 3 yield four testable hypotheses. Each compares a bond whose specialness exceeds the thresholds of some dealers with a bond at the GC rate.

H1. Depth at the best ask is lower, because fewer dealers quote at the best ask and those that remain post smaller quotes. Under the condition of Proposition 1(c), depth behind the best ask is not lower.

H2. Depth at the best bid is higher, because more dealers quote at the best bid and those already there post larger quotes. Under the condition of Proposition 1(c), depth behind the best bid is not higher.

H3. Aggressive sell orders are less frequent, and aggressive buy orders are not less frequent.

H4. The differences in H1 to H3 level off once specialness exceeds the thresholds of all dealers.

The model does not sign total depth on either side or the average size of the quotes at the best price. It predicts no change in the quoted spread or in the mid-quote as long as at least one dealer keeps quoting at each best price.

The static model says nothing about how long specialness lasts. In a dynamic version, the cost of selling would depend on whether the bond is hard to borrow at the coming settlement. The value of holding the bond would depend on whether it will still be hard to borrow at the later deliveries that it would cover. A change in specialness that does not last would therefore move the ask and aggressive selling more than the bid.

By Lemma 1, the cost works like a short position of k/γ, so the same predictions follow from anything else that raises dealers' reservation prices for a bond. Demand for the bond would do so, and so would actual short positions (Ho and Stoll, 1981; Guéant et al., 2013), which can also make the bond special. The tests can therefore show whether the book changes as the model predicts, but not that a sourcing cost causes the change. Two further implications narrow the alternatives. If short positions alone moved the book, repo borrowing above its recent average would reduce best-ask depth even for a bond at the GC rate. Without Assumption 1, the cost would arise only when a dealer must deliver more than it holds, so best-ask depth would fall more when the bond's repo borrowing is above average. Section 5.6 examines the timing, and Section 5.7 examines these implications and the most likely sources of demand.

4 Data and design

4.1 Data

The data cover Italian government bonds on MTS from 2 September 2019 to 26 February 2021. They contain all SC and GC spot-next trades on MTS Repo, the quotes and proposal records of the MTS cash order book, and the fill records of executed orders. I use spot-next repos because they settle two business days after the trade (Dufour et al., 2020), as cash trades do under the EU settlement rule (European Parliament and Council of the European Union, 2014). A dealer that sells a bond short can thus borrow it in spot-next repo for delivery. To clean the repo trades, I remove trades with missing rates or volumes and trades flagged as suspended. I also drop trades whose rate differs by more than 50 bp from the bond's previous trade on the same day. Of the remaining trades, I drop those whose rate lies more than five median absolute deviations plus 1 bp from the median of a window of 20 trades of the bond centred on them. I keep the bonds that trade in repo on at least 21 days. I exclude 20 February to 18 March 2020, the weeks of market turmoil before the announcement of the European Central Bank's pandemic emergency purchase programme (PEPP). This leaves a pre-COVID period from 2 September 2019 to 19

February 2020 and a post-PEPP period from 19 March 2020 to 26 February 2021. The regression sample has 126 bonds: 95 nominal BTPs (Buoni del Tesoro Poliennali), 12 BTP€i indexed to euro-area inflation, 12 floating-rate CCTs (Certificati di Credito del Tesoro) and 7 zero-coupon CTZs (Certificati del Tesoro Zero-coupon). It covers 98.76% of the cash-market bond-days with 15-minute quotes in the two periods (Appendix A2). All times are Central European Time.

4.2 Market making on MTS

MTS is the main interdealer market for Italian government bonds (Pelizzon et al., 2016). Market makers post two-sided quotes, called proposals, and other participants, including other market makers, trade against them with fill-and-kill orders. Market makers also have quoting obligations, and the monthly bond reference files of MTS record, for each bond, the number of market makers obliged to quote it, the largest quoted spread that an obliged quote may show and its minimum size. For the bonds in the sample, 6.17 market makers are obliged to quote a bond on average, between 4 and 14, and this number changes from one month to the next for 119 of the 126 bonds. An obliged quote must show at least €2 million, and the largest quoted spread it may show rises with residual maturity, from 0.05 to 2.00 points of price. The files also record a compliance time of five hours, the minimum time per trading day for which an obliged market maker must show a quote that meets these conditions. The obligations therefore bound the size and the quoted spread of a market maker's own quote, but they do not require the quote to stand at the best price. A market maker can meet its obligation with an ask behind the best ask, which is the choice that the model studies. The tick is 0.01 points of price, or 1 bp, for most bonds.

4.3 Variables

I index bonds by n and trading days by d. Specialness is the GC rate minus the bond's SC rate, in bp, with the GC rate measured at the time of the SC trade, because the GC rate moves within the day and SC trading concentrates in the morning. For bond n on day d,

$$S_{n,d} = \frac{\sum_k v_k \left(r^{\mathrm{GC}}_{h(k)} - r^{\mathrm{SC}}_k\right)}{\sum_k v_k}, \tag{6}$$

where the sums run over the SC spot-next trades k of the bond on that day, v_k is the volume of trade k and r^{SC}_k its rate, and $r^{\mathrm{GC}}_{h(k)}$ is the volume-weighted GC spot-next rate in the half hour of

the trade, or in the nearest half hour of the same day with GC trades. The median of specialness across bond-days is 2.04 bp, and specialness exceeds 2 bp on 51% of bond-days and is negative on 2% of them. Specialness measured from trades initiated by the borrower of the bond exceeds its value from trades initiated by the lender by 0.51 bp on average. The main results hold with the midpoint of the two and with lender-initiated trades only (Appendix A13).

For the best quotes, let $\mathrm{AP}_{n,d,i,t}$ and $\mathrm{BP}_{n,d,i,t}$ be the best ask and bid prices in second t of the 15-minute interval i, and $\mathrm{AS}_{n,d,i,t}$ and $\mathrm{BS}_{n,d,i,t}$ the quantities posted at them. Depth on each side and the proportional quoted spread of the interval are averages over its T seconds,

$$\begin{aligned} \mathrm{DEPTH}^{A}_{n,d,i} &= \frac{1}{T}\sum_{t=1}^{T} \mathrm{AS}_{n,d,i,t}, \qquad \mathrm{DEPTH}^{B}_{n,d,i} = \frac{1}{T}\sum_{t=1}^{T} \mathrm{BS}_{n,d,i,t}, \\ \mathrm{PQS}_{n,d,i} &= \frac{1}{T}\sum_{t=1}^{T} \frac{\mathrm{AP}_{n,d,i,t} - \mathrm{BP}_{n,d,i,t}}{(\mathrm{AP}_{n,d,i,t} + \mathrm{BP}_{n,d,i,t})/2}. \end{aligned} \tag{7}$$

I average best-ask depth DEPTH^{A}, best-bid depth DEPTH^{B} and the quoted spread over the intervals of the bond-day and use 100 times the log of the bond-day mean as the outcome. The ask/bid depth ratio is $100\ln(\mathrm{DEPTH}^{A}/\mathrm{DEPTH}^{B})$, and I call the proportional quoted spread the quoted spread.

The full order book comes from the exchange's proposal records, in which each two-sided proposal is active from its update to its end time. I rebuild the book of every bond every 15 minutes from 08:15 to 17:30 local time and measure depth at the best price, behind it and in total on each side. Its best quotes equal the 15-minute best quotes wherever the quotes do not move, once both use the same proposals (Appendix A2). Figure 1 shows these measures in the book of one bond at one snapshot. Executed aggressive orders come from the fill records, with time, bond, aggressor side and quantity, but they do not show whether a seller is short.

Table 1 summarises specialness, the repo controls and the book. Depth at the best ask and at the best bid is similar on average. The best price holds a little over a third of the depth on each side of the rebuilt book. As Table 2 shows, an executed order occurs on 67% of bond-days. On the bond-days with an executed sell, there are on average 3.77 executed sells, for €39.77 million. Bond-days above 2 bp of specialness have fewer executed sells and about as many executed buys as bond-days below.

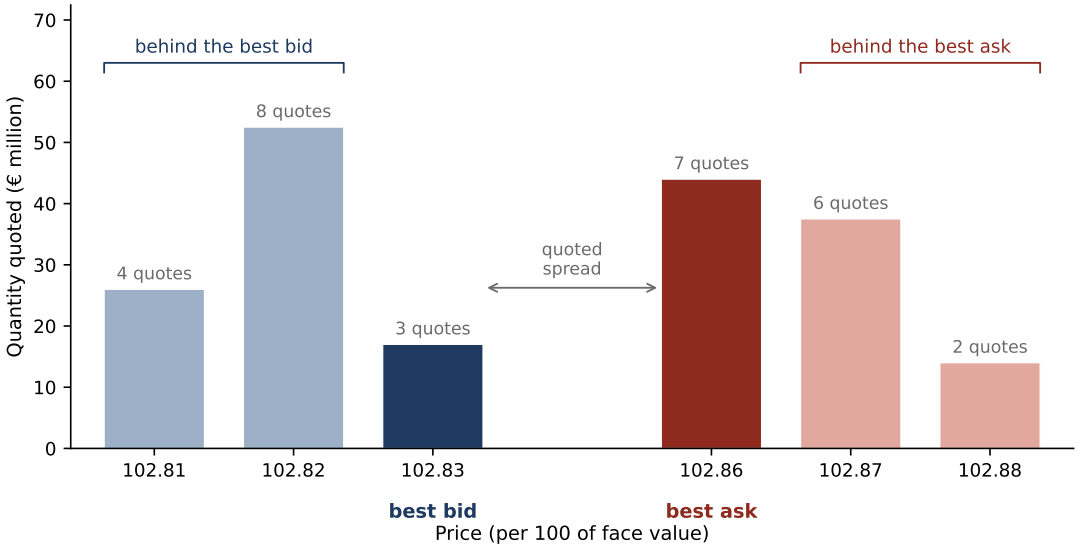


Figure 1: The order book of one bond. This figure shows the order book of the BTP 0.95% 1 March 2023 on MTS at 11:00 local time on 5 February 2020, a day on which the bond's specialness was 0.50 bp. The book is rebuilt from the proposal records as in Section 4.3 and holds 16 two-sided proposals. Each bar is the quantity quoted at one price, and the number above it counts the quotes at that price. The darker bars are depth at the best bid and at the best ask, €17 million and €44 million. The lighter bars are depth at the next two prices on each side. In all, the book holds €83.50 million behind the best bid and €56.50 million behind the best ask.

4.4 Specification

I estimate

$$Y_{n,d} = \alpha_n + \tau_d + \beta \tilde{S}_{n,d} + \lambda_- \min(S_{n,d}, 0) + \lambda_+ \max(S_{n,d} - 2, 0) + \mu_1 \bar{\sigma}_{n,d} + \mu_2 \bar{U}_{n,d} + \varepsilon_{n,d}. \quad (8)$$

Here $\tilde{S} = \min\{\max(S, 0), 2\}$ is capped specialness, which lies between zero and 2 bp. The outcome Y is a feature of the book or of the order flow, and α_n and τ_d are bond and day fixed effects. Because capped specialness is 0 for a bond at the GC rate and 2 for a bond with 2 bp of specialness, the difference between the two bonds is 2β. This piecewise-linear specification approximates Proposition 3: the outcome changes over the range of the thresholds and flattens beyond it, and the part below zero has its own slope. The slope changes at 2 bp, where the ask/bid depth ratio and selling flatten (Figure 3), and the slope above this breakpoint is estimated separately. With the breakpoint at 1 or 3 bp, the ratio and best-ask depth still fall significantly (Appendix A3). The control $\bar{\sigma}$ is the mean over the previous 20 trading days of the log intraday dispersion of the bond's SC rate (Schaffner et al., 2019). Dispersion is the realised volatility of the bond's SC rate path, whose value in each second with SC trades is their volume-weighted rate. It is the square root of the sum of the squared changes between

Table 1: Summary statistics. This table reports summary statistics for the pre-COVID and post-PEPP days. It uses the bond-days on which the past repo measures exist, which requires at least 10 of the previous 20 trading days. The last two columns are means over bond-days with specialness at or below 2 bp and above 2 bp. Panel A covers the bond-days with an SC repo path and 15-minute cash quotes. Specialness is the GC spot-next rate of the same half hour minus the bond's SC spot-next rate, volume-weighted over the bond-day (Section 4). Dispersion is the realised volatility of the bond's SC rate path at a one-second resolution, and rate updates count the changes of that path (Section 4). Depth is the mean quantity at the best quotes over the day. The quoted spread is the proportional quoted spread. Panel B covers the full order book rebuilt from the proposal records. The book is rebuilt every 15 minutes from 08:15 to 17:30 local time. The sample has 36,622 bond-days on 126 bonds over 346 days.

	All bond-days					Mean by specialness	
	Mean	SD	P10	Median	P90	≤2 bp	>2 bp
Panel A: Best quotes and repo (36,622 bond-days)							
Specialness (bp)	2.93	4.54	0.30	2.04	6.11	0.87	4.92
SC rate dispersion (bp)	3.19	3.53	0.87	2.55	5.94	1.87	4.48
SC rate updates (number)	18.19	8.78	8	17	29	14.54	21.74
Ask depth at the best quote (€ million)	45.64	47.13	8.78	33.14	92.61	47.18	44.15
Bid depth at the best quote (€ million)	46.64	50.82	8.84	34.16	92.50	45.76	47.50
Quoted spread (bp)	24.60	38.59	5.07	14.07	51.76	25.02	24.19
Panel B: Full order book (36,622 bond-days)							
Ask depth at the best price (€ million)	45.50	46.33	8.75	33.45	92.35	46.90	44.15
Bid depth at the best price (€ million)	46.44	50.61	8.64	34.00	92.35	45.58	47.28
Ask depth behind the best price (€ million)	61.29	24.72	35.47	58.06	87.39	62.39	60.23
Bid depth behind the best price (€ million)	60.50	24.36	35.76	57.22	85.64	63.06	58.00
Share of ask depth at the best price (%)	36.82	17.13	14.48	36.08	59.54	35.80	37.81
Share of bid depth at the best price (%)	37.34	17.42	14.51	36.57	60.64	34.52	40.08
Active two-sided quotes (number)	15.79	1.32	14.26	15.79	17.43	15.74	15.84

consecutive such seconds. The control $\bar{U}$ is the mean over the same days of the log number of these changes, the rate updates. I use past values because same-day repo activity responds to the day's trading. The controls matter for the size of the estimates. Special bonds have more SC rate updates, 21.74 a day against 14.54 (Table 1), and bonds with more updates have less depth at the best ask (Table 3). Appendix A6 reports the estimates without these controls. Standard errors are clustered by bond and by day (Cameron et al., 2011).

5 Results

5.1 The best quotes

Panel A of Table 3 shows that a bond with 2 bp of specialness has 7.99 log points less depth at the best ask than a bond at the GC rate, significant at the 1% level. The bond has 8.41 log points more depth at the best bid, significant at the 1% level. Its ask/bid depth ratio is 16.39 log points lower, or 0.39 times the ratio's standard deviation of 42 log points across bond-days. Its quoted spread does not differ significantly, as Proposition 1 allows. Figure 2 shows

Table 2: Executed orders. This table reports summary statistics for the executed fill-and-kill orders in the fill records, by bond-day. An executed order is counted once, whatever its number of fills. The first two columns use all bond-days. The third column is the mean over the bond-days with at least one executed order on that side. For the shares of bond-days with an order, the table reports only means. The last two columns are means over bond-days with specialness at or below 2 bp and above 2 bp. The sample has the 36,622 bond-days of Table 1.

	All bond-days			Mean by specialness	
	Mean	SD	Mean given an order	≤2 bp	>2 bp
Bond-days with an executed order (%)	67.43			68.88	66.01
Bond-days with an executed sell (%)	50.37			54.16	46.68
Bond-days with an executed buy (%)	47.12			46.26	47.97
Executed sell orders (number)	1.90	3.42	3.77	2.20	1.61
Executed buy orders (number)	1.65	3.12	3.51	1.64	1.67
Executed sell volume (€ million)	20.03	44.28	39.77	23.18	16.97
Executed buy volume (€ million)	17.20	38.92	36.50	16.50	17.89
Net executed buy volume (€ million)	−2.83	50.46		−6.68	0.92

the differences in depth across levels of specialness. Best-bid depth rises up to about 1 bp of specialness and changes little above 2 bp. Best-ask depth falls more gradually, and the estimates above 2 bp are imprecise (Section 5.5). In Panel B, best-ask depth is 7.16 log points lower at 2 bp than at the GC rate after the PEPP announcement, significant at the 5% level, and 5.75 log points lower before COVID, not significant at conventional levels. The difference between the two periods is not statistically significant (Appendix A9), and the earlier period has less than half as many bond-days.

How large must the sourcing cost be to produce these differences? In the model, a quote that stays at the best ask shrinks by k/γ, so its log falls by $\ln(m/(m-k))$, where m is the distance between the ask price and the dealer's reservation price. The model does not tie this distance to the quoted spread, and it differs across dealers, so the calculation is an illustration. I set it at half the median quoted spread, 7.04 bp of price (Table 1). If the whole fall of 7.99 log points came from smaller quotes, k would be 0.54 bp of price, or about €540 per €10 million. If only the part that Section 5.2 attributes to smaller quotes did, k would be 0.22 bp, or about €225, although that part is not significant on its own. A dealer that sells the bond short may have to roll the borrowing until it can buy the bond back. A bond above 2 bp stays above 2 bp on the next repo day with a probability of 90%, so a dealer can expect to roll the borrowing of €10 million for about 10 nights, at a cost of about €56. The implied cost is 4 to 10 times this amount and 40 to 97 times the cost of one night, €5.56 (Section 3). These ratios scale with the assumed distance and with how long a dealer stays short, so the calibration does not establish a cost beyond the repo rate, to which the costs of locating the bond and of failed deliveries

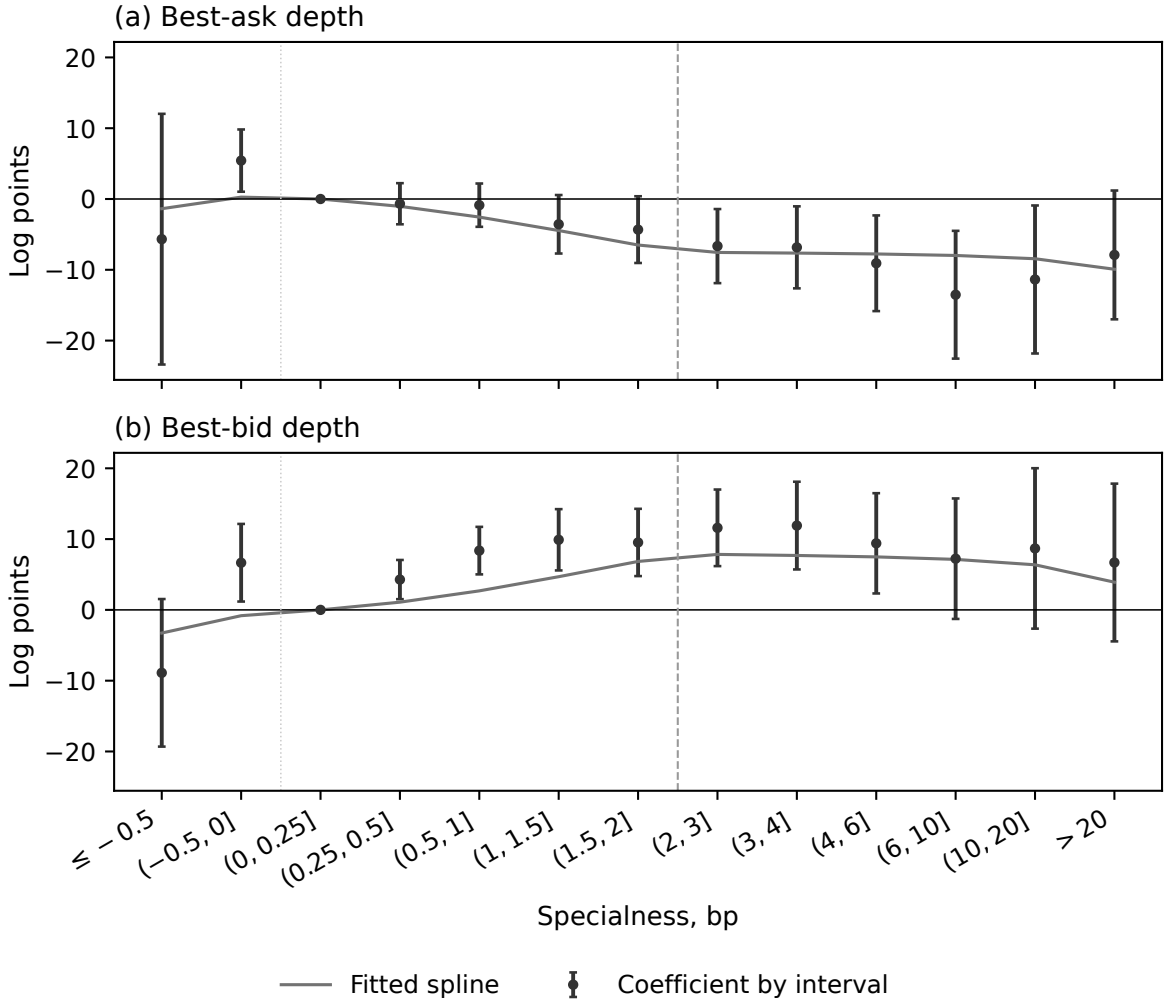


Figure 2: Depth at the best quotes across levels of specialness. Each panel plots coefficients on indicators for intervals of specialness, in bp, relative to the interval from 0 to 0.25 bp, with 95% confidence intervals. The indicators replace the three spline terms in equation (8). Each regression uses the 36,622 bond-days of Table 3, with bond and day fixed effects and the past dispersion and rate updates. Panel (a) uses depth at the best ask and panel (b) depth at the best bid, both 100 times their logs, on one scale. The line is the fitted spline from Table 3 at the median specialness of each interval. The dashed line marks 2 bp and the dotted line marks zero. A Wald test that the coefficients of the intervals above 2 bp are equal gives $p = 0.50$ for best-ask depth and $p = 0.34$ for best-bid depth.

would add. Dealers close to a cutoff move their quotes for any positive k, so the number of quotes that move depends on how many dealers stand near the cutoffs. The calibration uses proportional changes, because the changes in euros are imprecise when depth is skewed across bonds (Appendix A4). If both sides of a quote can execute, quotes change by less than k/γ, and the calibration understates k (Appendix A1).

Table 3: Specialness and the best quotes. Panel A reports estimates of equation (8) with bond and day fixed effects. Columns (1) to (4) use the bond-day means of the 15-minute best quotes. The ask/bid depth ratio is 100 times the log of ask depth over bid depth. Ask depth, bid depth and the proportional quoted spread are 100 times their logs. Column (5) is an indicator, in percentage points, for a bond-day with an executed sell order. Specialness enters in bp, and specialness from 0 to 2 bp is capped specialness. The first row reports 2β, where β is the coefficient on capped specialness. It is the difference between a bond at the GC rate and a bond with 2 bp of specialness. The parts of specialness below zero and above 2 bp enter separately, with coefficients per bp. The past dispersion and rate updates are the means over the previous 20 trading days of the log dispersion and of the log number of updates of the bond's SC rate, standardised. Panel B reports 2β with the controls of Panel A. The number of bond-days of each specification is under its name. The first three rows of Panel B replace the bond fixed effects with bond-by-quarter, bond-by-month or bond-by-week fixed effects. The next two rows estimate the model within each period. The last row regresses the same-day change in the outcome on the same-day change in capped specialness, so that 2β is the change on a day on which capped specialness rises from 0 to 2 bp. It uses consecutive observations of a bond at most four calendar days apart, with bond and day fixed effects and the past dispersion and rate updates. Appendix A6 reports the estimates without the repo controls. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Ask/bid depth ratio (1)	Best-ask depth (2)	Best-bid depth (3)	Quoted spread (4)	Any executed sell (5)
Panel A: Bond and day FE					
GC rate to 2 bp of specialness	-16.39^{***}	-7.99^{***}	8.41^{***}	1.35	-8.73^{***}
	(2.28)	(2.83)	(3.09)	(2.46)	(1.79)
Specialness below 0, per bp	-1.36	2.73	4.09	-6.14	-1.27
	(6.67)	(7.57)	(5.13)	(4.32)	(5.97)
Specialness above 2 bp, per bp	0.06	-0.09	-0.15	0.14	0.14
	(0.08)	(0.11)	(0.10)	(0.10)	(0.10)
Past SC rate dispersion	0.30	1.28	0.98	1.15	-1.90^{**}
	(1.25)	(2.33)	(2.34)	(1.58)	(0.91)
Past SC rate updates	-3.75^{**}	-9.77^{***}	-6.02^{**}	1.66	2.38^{**}
	(1.50)	(2.99)	(3.01)	(1.91)	(1.15)
Bond FE	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes
Observations	36,622	36,622	36,622	36,622	36,622
Panel B: GC rate to 2 bp of specialness in other specifications					
Bond × quarter FE	-12.47^{***}	-7.64^{***}	4.83^{**}	1.28	-3.90^{**}
36,622 bond-days	(1.93)	(1.86)	(1.94)	(1.86)	(1.71)
Bond × month FE	-8.86^{***}	-5.39^{***}	3.47^{**}	0.70	-0.22
36,622 bond-days	(1.70)	(1.53)	(1.65)	(1.73)	(1.91)
Bond × week FE	-4.05^{**}	-3.09^{*}	0.96	6.74^{***}	3.63
36,622 bond-days	(1.88)	(1.70)	(1.80)	(2.16)	(2.76)
Pre-COVID period	-13.79^{***}	-5.75	8.03^{**}	1.34	-7.22^{***}
10,897 bond-days	(4.24)	(3.50)	(3.60)	(3.63)	(2.35)
Post-PEPP period	-15.45^{***}	-7.16^{**}	8.30^{***}	0.56	-6.84^{***}
25,725 bond-days	(2.52)	(2.79)	(3.12)	(2.48)	(2.24)
Same-day change	-5.46^{***}	-3.10^{*}	2.36	6.59^{***}	2.81
36,067 bond-days	(1.72)	(1.59)	(1.83)	(2.29)	(2.69)

5.2 Depth at and behind the best price

Table 4 tests the predictions on the full order book. Ask depth at the best price falls by 7.59 log points, significant at the 1% level. It falls by 8.55 log points more than depth behind the best price. Depth behind the best ask does not change significantly, even at the second and third best prices, and neither does total ask depth. The fall is therefore concentrated at the best price, as H1 predicts, although the estimates do not show that the depth leaving the best ask reappears behind it (Appendix A4). On the bid side, depth at the best price rises by 7.72 log points, significant at the 5% level, and by 11.76 log points relative to depth behind it, significant at the 1% level. Depth behind the best bid falls by 4.03 log points, significant at the 1% level, and total bid depth does not change significantly, which is consistent with H2.

In the model, fewer quotes stand at the best ask and the quotes that stay there are smaller. The average size of the quotes at the best price is not signed, because the quotes that leave are the smallest. On every eighth trading day I count the quotes at the best prices (Appendix A8). Of the fall in best-ask depth, 3.38 log points come from fewer quotes and 3.25 from a smaller average size, but neither part is significant on its own. The share of snapshots with a single quote at the best ask rises by 2.56 percentage points, significant at the 5% level. At the best bid, almost all of the rise comes from more quotes, 8.75 log points, significant at the 1% level.

Quote moves within the day show how the book adjusts. On the same days, I follow each proposal from one 15-minute snapshot to the next. A quote moves away from the best ask when the dealer raises its own ask and the quote stands behind the best ask 15 minutes later. It moves to the best ask when the dealer lowers its own ask and the quote stands at the best ask. A change caused only by another dealer's quote does not count as a move. If dealers re-optimise their quotes from time to time, the model implies that the cost makes quotes join the best ask less often or leave it more often, and the reverse at the best bid. Table 5 shows that the adjustment runs mainly through the quotes that join the best price. From the GC rate to 2 bp, the share of quotes behind the best ask that move to it within 15 minutes falls by 0.54 percentage points, from a mean of 6.36%, significant at the 10% level. The share of quotes behind the best bid that move to it rises by 0.40 percentage points, from a mean of 6.46%, significant at the 10% level. Moves away from the best prices do not change significantly, and neither do the quotes that leave the book from the best ask or the new quotes there. At the best bid, fewer quotes leave the book, by 0.46 percentage points, significant at the 10% level. On the ask side, the adjustment thus runs through existing quotes, whereas on the bid side fewer quotes also leave

the book from the best price. In a steady state, the two rates imply that the share of quotes at the best ask falls by about 6.31 log points and the share at the best bid rises by about 7.01 log points. The bid figure is close to the count above, while the ask figure is about twice the count. These estimates rest on 42 days and are significant only at the 10% level.

Column (2) of Table 4 adds past signed order flow, the executed sells minus buys over the previous 5 and 20 days and since the start of the sample. Market-maker inventories affect liquidity (Comerton-Forde et al., 2010), and I use these sums as proxies for dealer inventory and for recent buying pressure. The estimates shrink by up to a third and keep their signs. The fall in best-ask depth is also similar whether or not dealers have absorbed net selling. With the 20-day proxy one standard deviation above its mean, best-ask depth falls by 6.23 log points from the GC rate to 2 bp. With the proxy one standard deviation below its mean, it falls by 7.62 log points, and the difference is not significant at conventional levels.

5.3 Executed orders

From the GC rate to 2 bp, the probability of an executed sell falls by 8.73 percentage points on a base of 50%, significant at the 1% level. The number of executed sells falls by 25% (Table 6), while executed buying rises. The probability of an executed buy rises by 3.64 percentage points, significant at the 5% level, and net executed buying rises by €9.17 million per bond-day. Selling falls by 12.25 percentage points relative to buying, significant at the 1% level, as H3 predicts (Panel C of Table 4). Sell orders in the order files, filled or not, fall about as much as executed ones, and their fill rate does not change significantly (Appendix A5). The fall thus reflects fewer sell orders. The rise in buying is also consistent with demand for the bond (Section 5.7).

5.4 The cost of immediacy

The differences in the book affect traders who need immediacy. At each 15-minute snapshot, I let an order of €10 or 25 million walk the rebuilt book level by level. A buy takes the quantity at the best ask, then at the next ask price, and so on until it is filled, and a sell does the same down the bid prices. The cost of a buy is the average price paid minus the mid-quote, and the cost of a sell is the mid-quote minus the average price received, both in bp of the mid-quote. On an average bond-day, the whole book cannot fill an order of €25 million in 1.97% of the

Table 4: Tests of the model's predictions. Panels A to C report the difference between a bond at the GC rate and a bond with 2 bp of specialness. This difference is 2β, with β from equation (8). Panels A and B use the full order book rebuilt from the proposal records. The book is rebuilt every 15 minutes from 08:15 to 17:30 local time. Panel C uses the executed orders from the fill records. Panel D reports the slope per bp between 2 and 20 bp. It comes from equation (8) with the part above 2 bp capped at 20 bp and an indicator for specialness above 20 bp. The ratio and ask depth in Panel D come from the 15-minute best quotes. The second column gives the sign of each difference that the model predicts, + for a rise, − for a fall and 0 for no change. A blank cell marks a sign that the model leaves open. The signs for depth behind the best price, and for depth at the best price minus behind it, assume the condition of Proposition 1(c). Column (2) adds past signed order flow, the executed sells minus buys over the previous 5 and 20 days and since the start of the sample, scaled by the bond's past average volume. These sums proxy for dealer inventory and for recent buying pressure. Both columns use the 35,737 bond-days on which they exist, so column (1) differs slightly from Tables 3 and 6. The main estimate for best-ask depth is the 7.99 log points of the 15-minute best quotes in Table 3. The full book gives 7.83 on all bond-days and 7.59 here. All regressions include bond and day fixed effects and the controls of equation (8). Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Predicted sign	Baseline (1)	With past order flow (2)
Panel A: Ask side, full order book, log points (H1)			
Depth at the best price	−	-7.59^{***}	-6.85^{**}
		(2.82)	(2.77)
Depth behind the best price	+ or 0	0.96	0.56
		(1.38)	(1.34)
At the best price minus behind it	−	-8.55^{***}	-7.40^{**}
		(3.05)	(2.99)
Total depth		−1.77	−2.04
		(1.59)	(1.53)
Panel B: Bid side, full order book, log points (H2)			
Depth at the best price	+	7.72^{**}	5.38^{*}
		(3.00)	(2.79)
Depth behind the best price	− or 0	-4.03^{***}	-3.70^{***}
		(1.26)	(1.24)
At the best price minus behind it	+	11.76^{***}	9.08^{***}
		(3.40)	(3.22)
Total depth		0.71	0.10
		(1.55)	(1.47)
Panel C: Executed orders, percentage points (H3)			
Any executed sell	−	-8.57^{***}	-7.05^{***}
		(1.78)	(1.69)
Any executed buy	+ or 0	3.68^{**}	2.56^{*}
		(1.48)	(1.46)
Sell minus buy	−	-12.25^{***}	-9.62^{***}
		(1.86)	(1.67)
Panel D: Slope per bp between 2 and 20 bp (H4)			
Ask/bid depth ratio	− or 0	−0.25	−0.18
		(0.32)	(0.29)
Ask depth at the best quote	− or 0	−0.71	−0.57
		(0.46)	(0.46)
Any executed sell	− or 0	0.15	0.16
		(0.24)	(0.24)

Table 5: Quote moves within the day. This table follows each proposal from one 15-minute snapshot to the next on every eighth trading day. A quote at the best ask moves behind it if the dealer raises its own ask and the quote stands behind the best ask 15 minutes later. A quote behind the best ask moves to it if the dealer lowers its own ask and the quote stands at the best ask. A change caused only by another dealer's quote is not a move. The bid side is the mirror image. Quotes that leave the book and new quotes at the best price are counted separately. Each outcome is the number of events over the number of quotes at, or behind, the best price, in per cent, summed over the pairs of snapshots of the bond-day. The first column is the mean. The second gives the sign that the model predicts if dealers re-optimise their quotes from time to time. A blank cell marks a sign that the model leaves open. The last column reports the difference between a bond at the GC rate and a bond with 2 bp of specialness, 2β with β from equation (8), in percentage points. The regressions have bond and day fixed effects and the controls of equation (8). A bond-day needs at least 10 quotes at and 10 behind the best price on the side, so the sample has 4,444 bond-days for the ask side and 4,445 for the bid side, on 42 days. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Mean (%)	Predicted sign	GC rate to 2 bp (pp)
Panel A: Ask side			
Quotes at the best ask that move behind it	11.83	+	0.08
			(0.41)
Quotes behind the best ask that move to it	6.36	−	−0.54*
			(0.27)
Quotes at the best ask that leave the book	7.76		0.23
			(0.26)
New quotes at the best ask	9.21		−0.09
			(0.29)
Panel B: Bid side			
Quotes at the best bid that move behind it	11.57	−	−0.58
			(0.50)
Quotes behind the best bid that move to it	6.46	+	0.40*
			(0.20)
Quotes at the best bid that leave the book	7.74		−0.46*
			(0.27)
New quotes at the best bid	9.35		−0.44
			(0.28)

snapshots on the ask side and in 1.96% on the bid side. I compute all costs, including the half quoted spread, on the snapshots in which the book can fill €25 million on both sides. Table 7 reports the difference between a bond at the GC rate and a bond with 2 bp of specialness. The clearest difference is where a large order is filled. A buy of €25 million is 2.65 percentage points less likely to be filled in full at the best ask, from a base of 52%. A sell of the same size is 3.33 percentage points more likely to be filled in full at the best bid, from a base of 53%. With bond-by-month fixed effects, the two differences are 1.30 and 1.21 percentage points, significant at the 5% and 10% levels. A buy of €25 million is also 0.17 percentage points less likely to be filled by the whole book, from a base of 98.03%, significant at the 10% level. Buying also costs more relative to selling: the gap rises by 0.08 bp for €10 million and by 0.14 bp for €25 million, about €340 per order, both significant at the 1% level. With bond-by-month fixed effects, the gaps are 0.04 bp, significant at the 10% level, and 0.05 bp, not significant at conventional levels.

Table 6: Executed orders by side. This table reports estimates of equation (8) on the executed-order panel from the fill records, with bond and day fixed effects. Columns (1) and (4) are indicators for an executed sell order and an executed buy order on the bond-day, in percentage points. Column (2) is a Poisson pseudo-maximum-likelihood regression of the number of executed sell orders. Its coefficients are semi-elasticities, multiplied by 100 to give log points, so that a difference of x log points is a change of $100(e^{x/100} - 1)\%$. Column (3) uses 100 times the log of executed sell volume on the bond-days with an executed sell. Column (5) uses executed buy volume minus executed sell volume, in € million. Specialness and the controls enter as in Table 3, and the first row reports 2β, the difference between a bond at the GC rate and a bond with 2 bp of specialness. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Any executed sell (1)	Sell orders (PPML) (2)	Sell volume given a sell (3)	Any executed buy (4)	Net buy volume (5)
GC rate to 2 bp of specialness	-8.73^{***}	-29.09^{***}	-18.74^{***}	3.64^{**}	9.17^{***}
	(1.79)	(7.15)	(4.58)	(1.44)	(1.80)
Specialness below 0, per bp	-1.27	-26.97	5.03	6.64	0.10
	(5.97)	(19.98)	(18.92)	(4.34)	(4.14)
Specialness above 2 bp, per bp	0.14	0.32	-0.25^{*}	0.16	0.26^{*}
	(0.10)	(0.21)	(0.15)	(0.15)	(0.14)
Past SC rate dispersion	-1.90^{**}	-7.03^{**}	2.25	-0.86	-0.49
	(0.91)	(3.37)	(2.84)	(1.08)	(0.83)
Past SC rate updates	2.38^{**}	7.87^{**}	-0.28	5.46^{***}	3.32^{***}
	(1.15)	(3.98)	(3.03)	(1.46)	(1.06)
Bond FE	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes
Observations	36,622	36,622	18,446	36,622	36,622

They are small next to the mean half quoted spread on these snapshots, 8.58 bp, but they fall on one side of the market, which the quoted spread does not show. This half quoted spread is below half the mean quoted spread of Table 1, which also covers books too thin to fill €25 million.

5.5 The shape of the response

Figure 3 shows that the ratio and selling fall much more slowly above about 2 bp. A Wald test does not reject that the coefficients of the intervals above 2 bp are equal, for the ratio or for selling. Between 2 and 20 bp the slopes are -0.25 log points and 0.15 percentage points per bp. Their 95% confidence intervals limit them to at most 11% and 7% of the slopes below 2 bp. The relation thus flattens sharply, as H4 predicts, although the data cannot show that it becomes exactly flat. For best-ask depth (Figure 2), the slope between 2 and 20 bp is -0.71 log points per bp, not significant at conventional levels, but its interval allows up to 44% of the slope below 2 bp in the same regression. A sourcing cost that keeps rising with specialness would produce such a pattern (Section 3.3). I therefore treat the breakpoint at 2 bp as a description of the data, not as a parameter of the model. The spline shows where the response flattens, but it

Table 7: The cost of immediacy. This table reports how the cost of an aggressive order in the rebuilt book varies with specialness. At each 15-minute snapshot, an order of €10 or 25 million walks the book level by level, from the best price outwards, until it is filled. The cost of a buy is the average price paid minus the mid-quote, and the cost of a sell is the mid-quote minus the average price received, both in bp of the mid-quote. The costs and the half quoted spread are bond-day means over the snapshots in which the whole book can fill an order of €25 million on both sides. The last four rows are the percentage of snapshots in which an order of €25 million can be filled in full at the best price and by the whole book. The first column is the sample mean. Columns (1) to (3) report the difference between a bond at the GC rate and a bond with 2 bp of specialness, 2β with β from equation (8). Column (1) has bond and day fixed effects and the controls of equation (8). Column (2) adds past signed order flow, as in Table 4. Column (3) replaces the bond fixed effects with bond-by-month fixed effects. The sample has the 36,746 bond-days with the rebuilt book and the controls, which differ slightly from Table 3. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Mean	Bond and day FE (1)	With past order flow (2)	Bond × month FE (3)
Half quoted spread (bp)	8.58	0.14	0.06	−0.14
		(0.43)	(0.44)	(0.28)
Cost of buying €10 million (bp)	8.81	0.19	0.09	−0.12
		(0.45)	(0.46)	(0.31)
Cost of selling €10 million (bp)	8.80	0.11	0.03	−0.17
		(0.45)	(0.46)	(0.28)
Buy minus sell, €10 million (bp)	0.01	0.08***	0.06***	0.04*
		(0.02)	(0.02)	(0.02)
Cost of buying €25 million (bp)	9.23	0.26	0.15	−0.14
		(0.48)	(0.49)	(0.37)
Cost of selling €25 million (bp)	9.20	0.12	0.05	−0.19
		(0.48)	(0.49)	(0.30)
Buy minus sell, €25 million (bp)	0.03	0.14***	0.11***	0.05
		(0.03)	(0.03)	(0.04)
€25 million buy filled at the best ask (pp)	51.99	−2.65**	−2.52**	−1.30**
		(1.28)	(1.26)	(0.63)
€25 million sell filled at the best bid (pp)	53.24	3.33***	2.33**	1.21*
		(1.24)	(1.16)	(0.67)
€25 million buy filled by the whole book (pp)	98.03	−0.17*	−0.17*	−0.07
		(0.08)	(0.09)	(0.08)
€25 million sell filled by the whole book (pp)	98.04	−0.08	−0.10	−0.01
		(0.10)	(0.11)	(0.09)
Bond-days		36,746	35,737	36,746

cannot recover the distribution of the dealers' thresholds. Below zero, where the model implies no response, the slope of the ratio is not significant (Appendix A9).

5.6 Frequency and timing

Panel B of Table 3 adds more restrictive fixed effects. With bond-by-month fixed effects, best-ask depth is 5.39 log points lower at 2 bp of specialness than at the GC rate, significant at the 1% level, and with bond-by-week fixed effects it is 3.09 log points lower, significant at the 10% level. The ask/bid depth ratio also falls within bond-weeks, by 4.05 log points, significant at

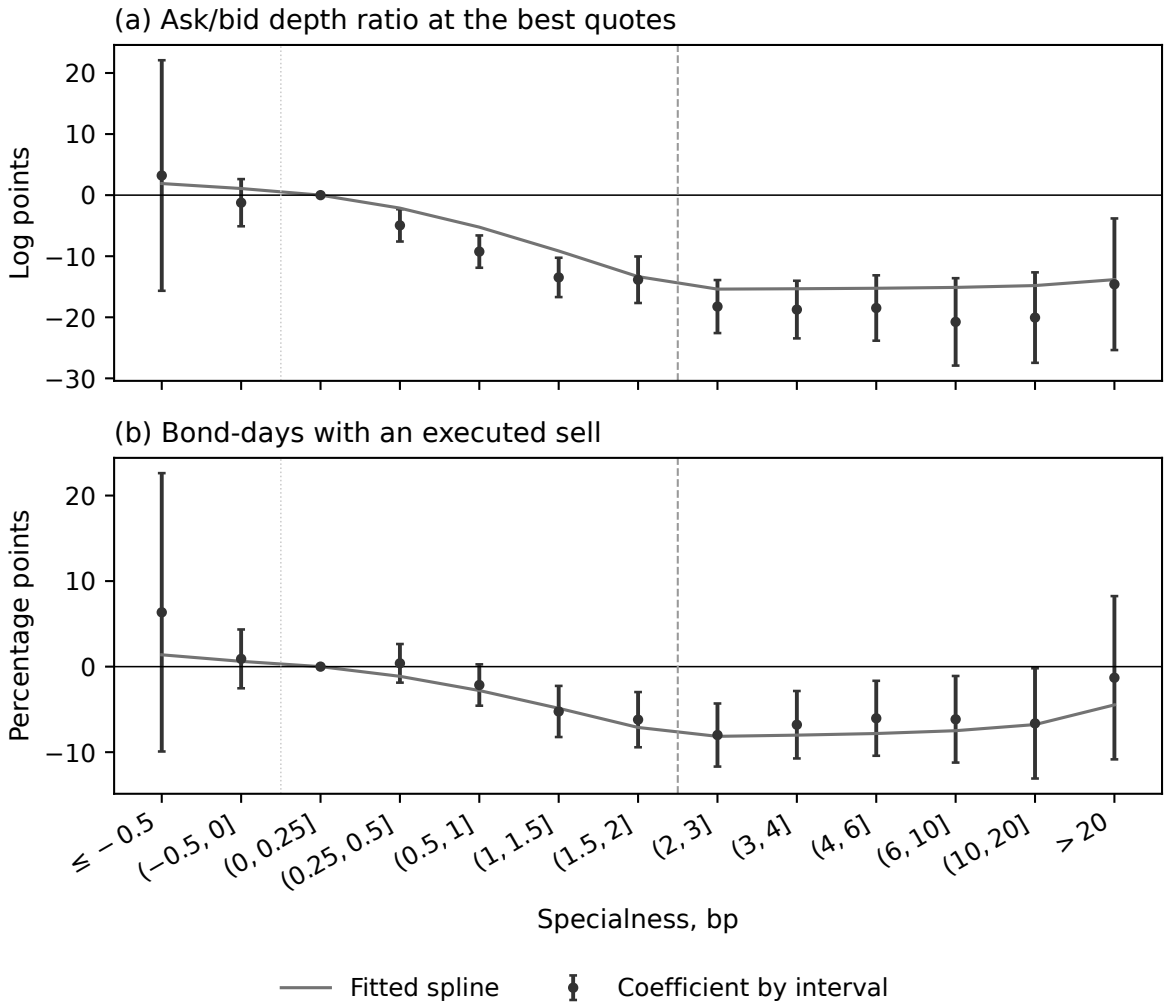


Figure 3: The ask/bid depth ratio and executed selling across levels of specialness. Each panel plots coefficients on indicators for intervals of specialness, in bp, relative to the interval from 0 to 0.25 bp, with 95% confidence intervals. The indicators replace the three spline terms in equation (8). Each regression uses the 36,622 bond-days of Table 3, with bond and day fixed effects and the past dispersion and rate updates. Panel (a) uses the ask/bid depth ratio at the best quotes, in log points. Panel (b) uses an indicator for a bond-day with an executed sell order, in percentage points. The line is the fitted spline from Table 3 at the median specialness of each interval. The dashed line marks 2 bp and the dotted line marks zero. A Wald test that the coefficients of the intervals above 2 bp are equal gives $p = 0.67$ for the ratio and $p = 0.50$ for selling.

the 5% level. On a day on which a bond's capped specialness rises from 0 to 2 bp, the ratio changes by −5.46 log points, significant at the 1% level, and best-ask depth by −3.10, significant at the 10% level. At these frequencies the quoted spread widens, by 6.74 log points from the GC rate to 2 bp with bond-by-week fixed effects, significant at the 1% level. In the model, the best ask moves only once every dealer quoting there bears the cost (Proposition 1(d)), so the model does not predict a wider quoted spread at moderate specialness, and the evidence on the quoted spread is mixed. It does not differ with bond and day fixed effects or within bond-months. It widens with supply-type daily changes in specialness, narrows with demand-type weekly changes

(Appendix A10) and narrows in the week after an entry into specialness (Section 5.7). The bid result holds with bond-by-month fixed effects, at 3.47 log points, significant at the 5% level, but not with bond-by-week fixed effects. Selling is associated with specialness only across months: with bond-by-week fixed effects and on the day capped specialness rises, its coefficients are positive and not significant.

The static model does not predict that the ask, the bid and selling differ at different horizons. The dynamic interpretation of Section 3.4 could account for the bid, because buying a bond to hold it pays only if specialness lasts, whereas selling it costs more whenever the bond is hard to borrow at the coming settlement. On that interpretation, however, selling should respond like the ask, and it does not: within bond-months and bond-weeks, its estimates are close to zero and exclude a response of the baseline size.

Specialness could respond to the book rather than the other way round. Dufour et al. (2020) find that specialness is higher after days with less quantity at the best ask prices. Table 8 therefore measures specialness from the SC and GC trades before 10:00 and the book and the executed orders from 10:00 on. Special spot-next trading starts at about 08:13 local time, and 64% of the day's SC volume has traded by 10:00. With morning specialness, best-ask depth after 10:00 falls by 7.07 log points from the GC rate to 2 bp, significant at the 1% level. This is about as much as with the whole day's specialness and the same outcomes from 10:00 (column (2) of Table 8). With best-ask depth before 10:00 as a control, it falls by 5.10 log points, significant at the 5% level. The ratio falls by 8.94 log points with the ratio before 10:00 as a control, significant at the 1% level. The incidence of an executed sell falls by 8.65 percentage points with morning specialness. In changes, however, a change in morning specialness does not move best-ask depth after 10:00 significantly (Appendix A10), so the evidence on timing within the day is inconclusive. Timing alone does not rule out news that moves both the repo rate and the book. Feedback from order flow to specialness across days is addressed in Section 5.7.

5.7 Alternative explanations

Three alternatives that raise dealers' reservation prices for a bond (Section 3.4) are consistent with some of these patterns. First, bonds that the Eurosystem buys become scarce in repo and in demand (Arrata et al., 2020). Second, primary dealers sell a bond short before it is reopened at auction, and its specialness rises from the announcement of the reopening until

Table 8: Specialness before 10:00 and the book from 10:00. Each cell is the difference between a bond at the GC rate and a bond with 2 bp of specialness, 2β with β from equation (8). The regressions include bond and day fixed effects and the parts of specialness below zero and above 2 bp. Column (1) uses whole-day outcomes and the day's specialness. Column (2) uses outcomes from 10:00 on, from the 15-minute best quotes, the full order book and the fill records, with the day's specialness. Column (3) uses the same outcomes with specialness from the SC and GC spot-next trades before 10:00 local time, matched by half hour as in Section 4. Columns (1) to (3) and (5) include the past dispersion and rate updates. Column (4) repeats column (3) without them. Column (5) adds the same outcome measured before 10:00. All columns use the same bond-days, those with SC and GC trades before 10:00. The depth imbalance is bid depth minus ask depth as a percentage of their sum. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

Outcome measured	Whole day	From 10:00	From 10:00	From 10:00	From 10:00
Specialness measured	Whole day	Whole day	Before 10:00	Before 10:00	Before 10:00
Past repo controls	Yes	Yes	Yes	No	Yes
Outcome before 10:00	No	No	No	No	Yes
	(1)	(2)	(3)	(4)	(5)
Ask/bid depth ratio, best quotes	−16.45***	−16.45***	−14.90***	−19.07***	−8.94***
	(2.39)	(2.68)	(2.50)	(2.63)	(1.91)
Ask depth at the best quote	−7.87***	−7.53***	−7.07***	−14.40***	−5.10**
	(2.77)	(2.60)	(2.46)	(3.00)	(1.96)
Bid depth at the best quote	8.58***	8.92***	7.83***	4.67	6.11***
	(3.09)	(2.98)	(2.85)	(3.14)	(2.34)
Depth imbalance (pp)	7.50***	7.65***	6.98***	9.00***	3.53***
	(1.11)	(1.20)	(1.13)	(1.18)	(0.82)
Ask depth at the best price, full book	−7.72***	−7.06***	−6.61***	−13.50***	−4.71**
	(2.76)	(2.64)	(2.48)	(2.91)	(1.99)
Bid depth at the best price, full book	8.64***	8.90***	7.95***	4.86	6.03**
	(3.09)	(3.01)	(2.89)	(3.17)	(2.36)
Bid depth behind the best price	−3.98***	−4.00***	−4.48***	−7.12***	−3.64***
	(1.32)	(1.38)	(1.28)	(1.48)	(1.12)
Bid depth, best minus behind	12.62***	12.90***	12.43***	11.98***	9.16***
	(3.57)	(3.61)	(3.49)	(3.84)	(2.74)
Any executed sell (pp)	−8.69***	−8.99***	−8.65***	−8.59***	−8.19***
	(1.85)	(1.76)	(1.70)	(1.68)	(1.65)
Any executed buy (pp)	2.83*	2.04	1.50	5.44***	1.08
	(1.53)	(1.56)	(1.52)	(1.50)	(1.48)
Bond-days			33,355		

shortly before settlement (Dufour et al., 2020). In the US Treasury market, prices and special repo rates also fall before auctions, which Lou et al. (2013) attribute to hedging sales by primary dealers. The cheapest-to-deliver bonds of the Euro-BTP futures on Eurex can also become special before delivery (Buraschi and Menini, 2002). Third, the value of a bond rises with its expected specialness (Duffie, 1996).

I collect the Treasury's auction announcements from the Bank of Italy and drop each auctioned bond from announcement to settlement, 1.73% of bond-days. From the GC rate to 2 bp, best-ask depth then falls by 8.07 log points, significant at the 1% level. It falls by 9.00 log points without the futures delivery windows, significant at the 1% level, and by 9.86 log points with the repo controls fixed before the bond's current special spell, significant at the 1% level. The same-day result also holds with the previous day's change in the outcome and the previous day's order flow as controls, at −3.79 log points for a rise from 0 to 2 bp, significant at the 1%

level (Appendix A9).

The ask result also holds without recent buying on MTS. On bond-days without net executed buying over the previous five days, best-ask depth falls by 8.43 log points from the GC rate to 2 bp. After net buying, it falls by 3.20 log points, and the difference is significant at the 5% level (Appendix A9). On a day on which capped specialness rises from 0 to 2 bp, the bond's closing mid-quote changes by −0.60 bp of price, and its 95% confidence interval rules out rises larger than 1.21 bp. These checks do not cover demand outside MTS, such as Eurosystem purchases, for which I lack holdings by bond.

Slow-moving changes within a bond could also move both its specialness and its book. A bond's age, its reopenings and whether it is the newest issue of its kind change its supply and its use as collateral, and the market makers obliged to quote it change from month to month. With controls for these characteristics, the ask/bid depth ratio falls by 15.23 log points from the GC rate to 2 bp and best-ask depth by 10.27, both significant at the 1% level. The rise in best-bid depth, 4.95 log points, is not significant at conventional levels. Within bond-months and with the same controls, they fall by 8.21 and 4.66 log points, both significant at the 1% level, and best-bid depth rises by 3.55, significant at the 5% level. Executed selling falls with the controls, but not within bond-months (Appendix A9).

Demand for the bond or dealers' short positions could also come first and make the bond special. A stacked event study of entries into specialness examines the order of events (Appendix A11). An entry is a bond-day with specialness above 2 bp after five consecutive observations of the bond at or below 1 bp within 14 calendar days, with specialness above 1 bp on at least two of the next three observations. On the entry day and the two days after it, specialness is 1.58 bp higher and best-ask depth 24.92 log points lower than ten to six days before, relative to bonds with stable specialness. This fall is significant at the 1% level with a wild-cluster bootstrap by bond. The event study has three limitations. First, the other outcomes do not follow the model at entry. Best-bid depth is 14.49 log points lower, which is not significant, and the ratio does not change significantly. In the following week, executed selling rises and the quoted spread narrows, while the fall in best-ask depth fades although specialness stays about 1 bp higher. Second, four rules with a jump above 1.5 bp after at most 1 bp give 77 to 229 entries. Under these rules, best-ask depth already changes by −7.98 to −14.93 log points in the five days before the entry, or 44 to 63% of the change on entry. This is the pattern expected if the book moves first, as Dufour et al. (2020) find, or if a common shock moves both. Third, the main rule

gives only 26 entries, 22 of them with a control bond. The timing evidence therefore shows a fall in best-ask depth around entries into specialness, but not that the repo market moves first.

Specialness could also proxy for dealers' short positions. Appendix A12 tests the first implication of Section 3.4 by splitting bond-days by specialness and by repo borrowing. Repo borrowing is above average when the bond's borrower-initiated SC volume exceeds its mean over the previous 20 trading days. Relative to a bond at the GC rate with below-average borrowing, best-ask depth differs by −8.87 log points for a special bond with above-average borrowing and by −6.15 for a special bond with below-average borrowing, both significant at the 1% level. For a bond at the GC rate with above-average borrowing, it differs by −1.54, which is not significant at conventional levels. Within bond-months and with the life-cycle controls, best-ask depth falls by 5.86 log points from the GC rate to 2 bp on bond-days with above-average borrowing and by 4.48 on the others, both significant at the 1% level. Repo borrowing here measures the flow of borrower-initiated trades on MTS, not the dealers' positions, and a special bond with below-average borrowing may also be held by a buy-and-hold owner such as the Eurosystem. The second implication concerns the difference between the specialness slopes of the two borrowing states. With bond and day fixed effects, this difference is −1.67 log points, with a standard error of 1.68, and within bond-months it is −1.38 (standard error 1.18). Both are consistent with Assumption 1 but do not separate the cost from demand for the bond or from the value of holding it. In changes, best-ask depth falls mainly when specialness and SC repo volume move in the same direction, the demand-type changes in the split of Cohen et al. (2007) for equity lending (Appendix A10). Neither split isolates a cost that arises from a scarce supply of the bond.

6 Conclusion

In Italian government bonds on MTS, repo specialness is associated with less depth at the best ask, with the difference concentrated at the best price. The association holds within bond-months, with controls for the bond's life cycle and with specialness cleaned of the repo bid-ask bounce. Special bonds have less best-ask depth whether their repo borrowing is above or below its recent average. Over longer horizons, special bonds also have more depth at the best bid and fewer executed sells relative to buys, and large buys are less likely to be filled in full at their best ask. Within the day, fewer quotes move to the best ask of a special bond and more

move to its best bid. The quoted spread widens within bond-weeks but narrows after entries into specialness, and the mid-quote does not change significantly.

The results have two implications. First, liquidity measures based on the quoted spread or on total depth can miss differences in liquidity that are associated with scarcity in repo, because these differences show up in depth at the best price on each side separately. Second, if specialness does shape the book, policies that change the supply of particular bonds, such as central bank purchases and securities lending facilities, could affect the two sides of the cash market differently.

A model in which a dealer must source a bond that is hard to borrow at a cost can account for the main differences with bond and day fixed effects. The cost raises the dealer's reservation price, much as a short position does in the standard inventory model, and with discrete prices dealers then withdraw from the best ask and join the best bid while the best prices stay where they are. The evidence does not identify that cost. Under the entry rules with a jump above 1.5 bp after at most 1 bp, best-ask depth starts to fall before the bond becomes special, and demand for the bond, including Eurosystem purchases, would produce the same patterns. Data on dealer positions and on Eurosystem holdings by bond would help to separate these explanations.

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